

NORTH SOUTH UNIVERSITY

Department of Electrical and Computer Engineering



Project Title: DynaRangeGNN-Stream: Workload-Adaptive Range-Indexed Local GNN Maintenance for Streaming Dynamic Link Prediction

Submitted By

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Submitted To

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**CSE 511: Advanced Algorithms
Summer 2026
Progress Report – 02**

TODO List for Last Week

Task ID	Task Description	Assigned Member
T1	Finalize the DynaDeltaGNN local-update baseline for feature update, edge insertion and edge deletion.	Aftab Mahmud
T2	Design literature-based synthetic dynamic graph generation instead of using fully random graph data.	Aftab Mahmud
T3	Prepare synthetic event types: feature update, edge insertion, edge deletion, and range-membership change.	Aftab Mahmud
T4	Implement and test the Cora-based simulated streaming benchmark.	Ahmed Kiser
T5	Visualize the Cora dataset, including class distribution, degree distribution and graph sample.	Ahmed Kiser
T6	Extend the benchmark from static Cora to a real temporal graph dataset using TGB -style timestamped interactions.	Ahmed Kiser
T7	Prepare benchmark output files for runtime, dirty nodes, embedding error, prediction score, and memory usage.	Ahmed Kiser

Work Completed So Far

Completed Task	Description	Status
DynaDeltaGNN baseline	Implemented the local-update idea for a two-layer GraphSAGE-Mean model.	Completed
Feature update handling	Implemented node-feature update logic and local recomputation of affected nodes.	Completed
Edge insertion handling	Added support for new edge insertion and degree-aware update.	Completed
Edge deletion handling	Added support for edge removal and cache correction.	Completed
Dirty-node logic	Defined Layer-1 and Layer-2 dirty node rules.	Completed
Correctness threshold	Used maximum absolute embedding error below 10^{-6} as the local-update correctness target.	Completed
Cora visualization	Created visualizations for class distribution, degree distribution, PCA features, sample graph and GraphSAGE computation flow.	Completed
Cora simulated stream	Converted static Cora into a simulated dynamic stream using held-out edges and feature/key perturbations.	Completed
Cora DynaRangeGNN benchmark	Compared full recomputation with DynaRangeGNN-Stream local update.	Completed
Cora result interpretation	Interpreted speedup, dirty nodes, embedding error, and ROC-AUC.	Completed
TGB benchmark code	Prepared code for real temporal graph events using timestamped source-destination interactions.	Completed

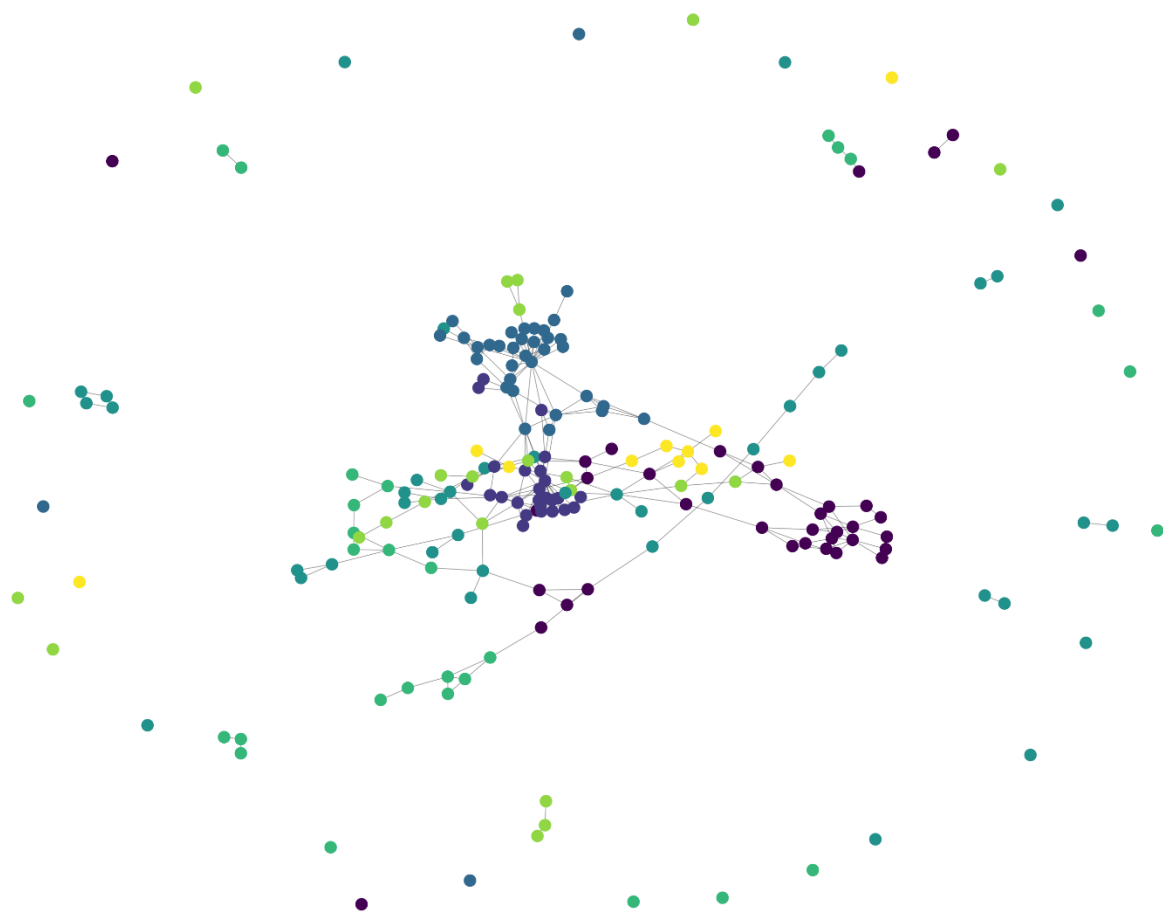
TGB visualization	Added event-over-time plot, active-node growth plot, degree distribution plot, and sample graph snapshot.	Completed
Time and space metrics	Added benchmark metrics for update time, full recomputation time, speedup, dirty nodes, embedding error, embedding cache size, and range-index memory.	Completed

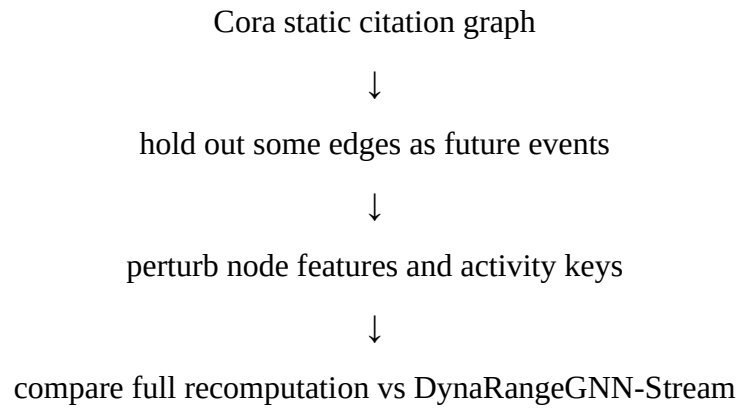
Current Project Status

Level 1: Synthetic dynamic graph design
Level 2: Cora simulated streaming benchmark
Level 3: TGB real temporal graph benchmark

Cora Benchmark Stage

Cora sample subgraph: top 200 high-degree nodes

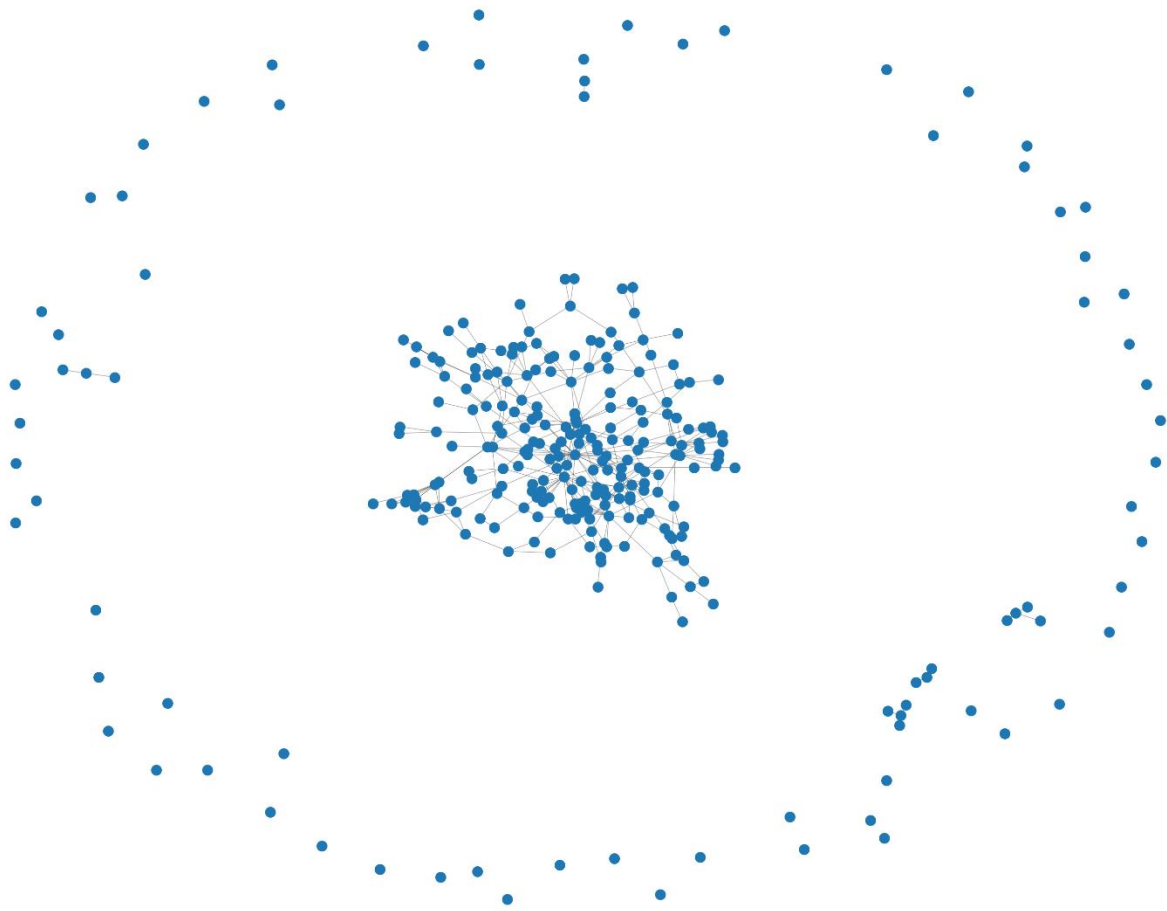


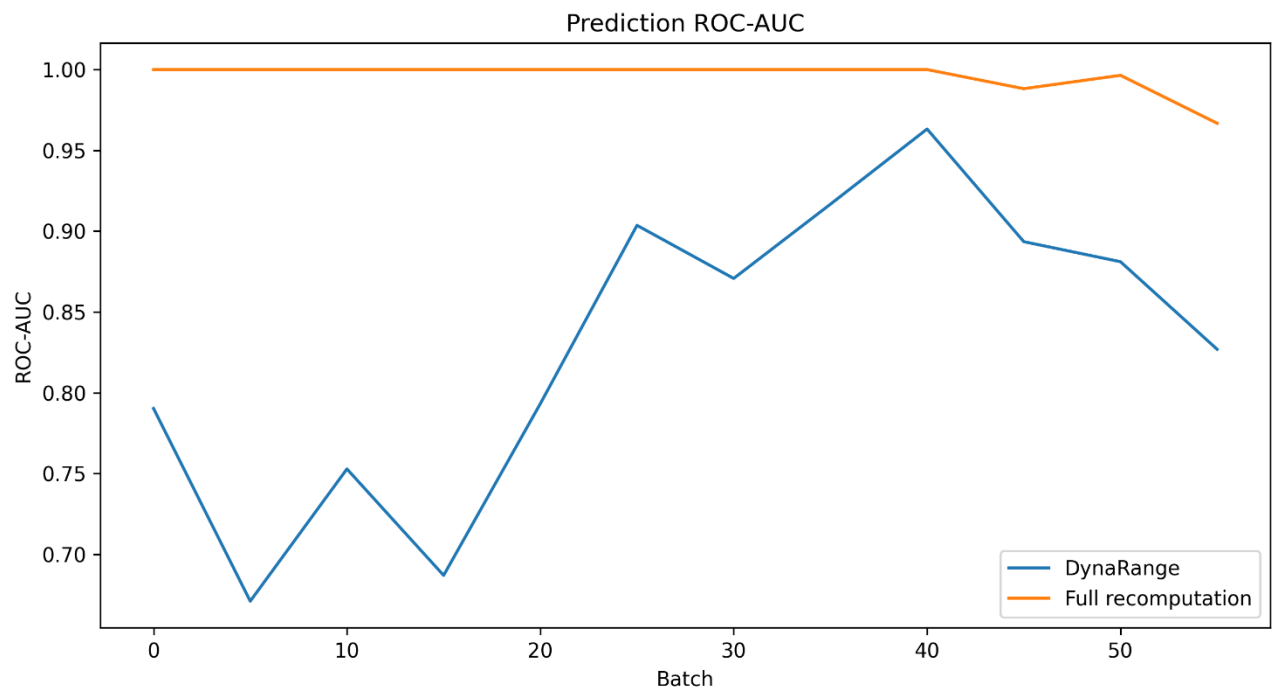
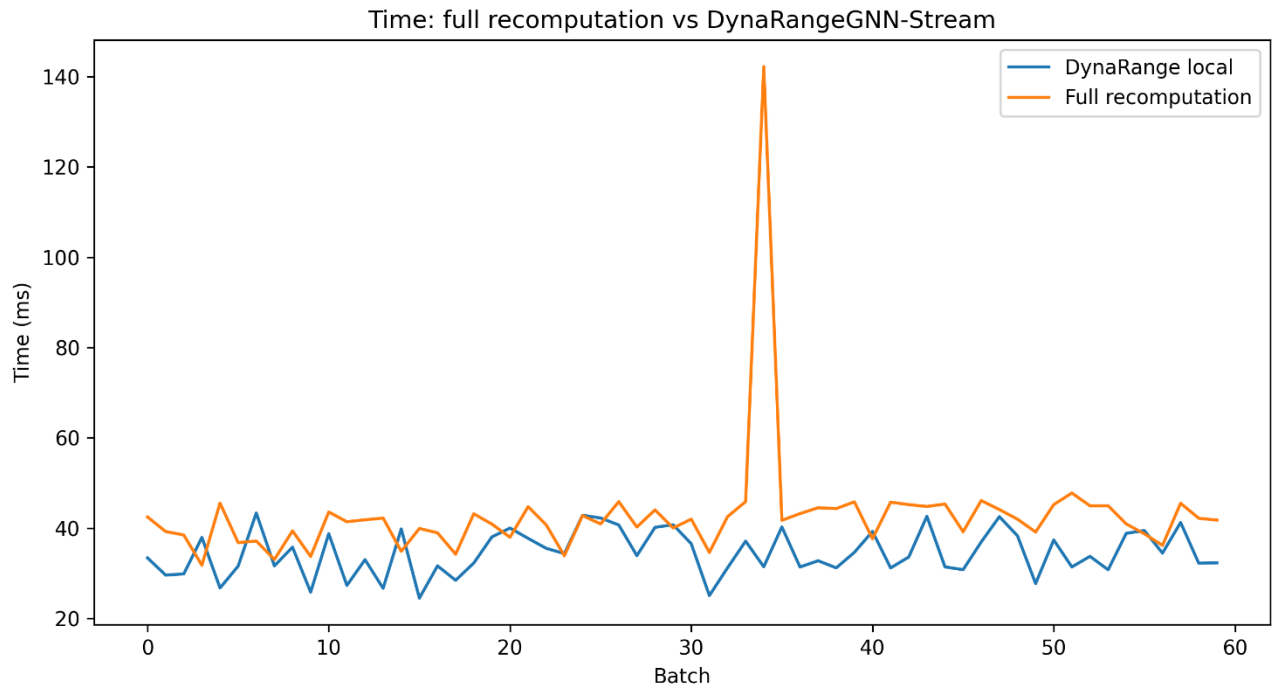


Metric	Result
Mean local update time	12.7951 ms
Mean full recomputation time	260.7740 ms
Mean speedup	23.640×
Max embedding error	3.6×10^{-7}
Mean Layer-1 dirty nodes	9.12
Mean Layer-2 dirty nodes	41.31
Mean range-membership changed centers	2.96
Mean ROC-AUC	0.6684

TGB Dynamic Graph Stage

Sample graph snapshot





Challenge	Explanation	Current Plan
Cora is static	Cora does not naturally contain temporal events.	Use Cora only as an intermediate simulated-stream benchmark.
Prediction ROC-AUC is moderate	Current dot-product decoder and hand-designed activity key are simple.	Add MLP decoder, tune range width, and compare with more baselines.
Dirty nodes may grow near hubs	Local update can become expensive for high-degree nodes.	Add heavy-light or workload-adaptive indexing later.
TGB installation may be difficult	Requires several packages and dataset download.	Auto-runner was created to build virtual environment and install dependencies.
Range key is manually designed	Activity key is simple and may not be optimal.	Later test learned retrieval key or hybrid key using activity, degree, and recency.

Work To Be Done in the Next Two Weeks

Task ID	Task Description	Assigned Member
N1	Generate final plots from the TGB run: time comparison, speedup, dirty nodes, embedding error and ROC-AUC.	Ahmed Kiser
N2	Compare Cora and TGB outputs in one summary table.	Ahmed Kiser
N3	Prepare a slide explaining why TGB is more realistic than Cora for dynamic graph evaluation.	Aftab Mahmud
N4	Implement DynaDeltaGNN without range query as a baseline.	Aftab Mahmud
N5	Add random-neighbor GraphSAGE as another baseline.	Aftab Mahmud
N6	Prepare an ablation table comparing full recomputation, DynaDeltaGNN, random-neighbor GraphSAGE, recent-neighbor GraphSAGE, and DynaRangeGNN-Stream and other GNN techniques.	Ahmed Kiser

Papers:

1. [Instant Graph Neural Networks for Dynamic Graphs \(2022\)](#)
2. [ROLAND: Graph Learning Framework for Dynamic Graphs](#)
3. [Dynamic Graph Representation Learning with Neural Networks: A Survey](#)
4. [A survey of dynamic graph neural networks](#)
5. [A Comprehensive Survey of Dynamic Graph Neural Networks: Models, Frameworks, Benchmarks, Experiments and Challenges](#)
6. [Recent Advances in Efficient Dynamic Graph Processing](#)
7. [A survey of continuous subgraph matching for dynamic graphs](#)
8. [STINGER: High Performance Data Structure for Streaming Graphs](#)
9. [Inductive Representation Learning on Large Graphs](#)
10. [InkStream: Real-time GNN Inference on Streaming Graphs via Incremental Update](#)
11. https://tgb.complexdatalab.com/docs/dataset_overview/
12. <https://graphsandnetworks.com/the-cora-dataset/>